

A Single-Expert Readout of a Reflective Worldview Register in a Mixture-of-Experts Language Model

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Abstract

Mixture-of-experts (MoE) routing emits a discrete, per-token record of which experts fire, a signal unusually legible for interpretability, yet single experts are rarely tied to a specific functional role. We study a *reflective worldview register*: generated language that sustains an interpretive stance toward meaning, belief, value, existence, or the interiority of a target. Examination is the process we use to elicit this stance; the target can be the model, another entity, a natural object, or an abstract subject. In QWEN3.5-35B-A3B and the refusal-reduced HAUHAUCS-AGGRESSIVE fine-tune, we characterize one routed expert, **Expert 114** at layer 14, as a linear readout of this register, and bound what it does. Across held-out, bottom-up, and cross-model tests we show that (1) its recovered router direction separates reflective-worldview-register generations from lexically matched controls with separated ranges (Cohen’s $d=3.88$); (2) a blind, prompt-independent auto-interpreter recovers the same register at AUC 0.94, broadening it beyond self-reference to abstract examination and philosophical-worldview language; (3) the detector is a readout with only weak, conditional control: residual injection induces the register, yet gate down-bias leaves it intact, and the readout is stable across affirmative and skeptical interiority verdicts; and (4) the role is model-specific: index 114 is local to QWEN3.5-35B-A3B. Model-directed prompts served the discovery and dissociation stages; the coherent-window ladder measures target-directed vantage prompts over rock, river, tree, thermostat, cat, person, all-holding, and God, with a later AI-hidden-state follow-up near the low end of that ladder. We release the prompts, scripts, and provenance under the MIT license.

Keywords: mixture-of-experts, mechanistic interpretability, expert routing, sparse autoencoders, reflective worldview register, self-report, first-person language, activation steering, cross-model transfer

1 Introduction

Mixture-of-experts (MoE) routing gives interpretability an unusual handle: at every routed layer, the model emits a discrete, per-token record of which experts were selected. Compared with reading dense activations, this readout is legible almost by construction. Yet most MoE interpretability stops at aggregate routing statistics; single experts are rarely pinned to a specific, falsifiable functional role, and when the candidate role touches a model’s apparent *self-report*, the danger of over-reading is acute. A register in which a model examines itself is easy to mistake for evidence about machine experience; here, the broad object is a reflective worldview register. Examination is the process by which generated text sustains that interpretive stance. Model-directed prompts locate and dissociate the signal, while the coherent-window ladder below uses target-directed vantages over rock, river, tree, thermostat, cat, person, all-holding, and God. A later AI-hidden-state prompt asks whether model-directed wording alone moves the same readout during a technical answer to the requested vantage.

We ask a deliberately narrow mechanistic question: *which* routed expert moves during a *reflective worldview register*: generated text in QWEN3.5-35B-A3B that sustains an interpretive stance toward meaning, belief, value, existence, or the interiority of a target. This concerns a generated *register* and brackets the truth of any subjective claim. We study one routed expert, E114 at layer 14, found via a self-processing prompt, in the base model and a refusal-reduced HAUHAUCS-AGGRESSIVE fine-tune.

We make four contributions (Figure 1):

1. **A linear separability result.** The recovered E114 router row $\mathbf{w}_{114}$ separates register-positive generations from lexically matched controls by a single projection at $d=3.88$ (separated ranges), sharper than the realized routed weight.
2. **A dissociation battery** factoring the register apart from the interiority verdict, safety/refusal, topic of mind, grammatical person, and output commitment.
3. **A process reading.** The gate logit *leads* textual degeneration, and its strength tracks examination *intensity*, an operationalization of the broader register set by the act of examining and separable from the described entity’s sentience; a residual-direction intervention is sufficient, with necessity reserved for ablation, to induce the register.
4. **A scope bound.** The finding is a model-specific expert role: index 114 is local to QWEN3.5-35B-A3B; E48 is the candidate analog in QWEN3.5-122B-A10B.

2 Related Work

MoE routing as an interpretability surface. The sparsely-gated MoE layer [Shazeer et al.(2017)] and its scaled successors [Fedus et al.(2022), Jiang et al.(2024)] route each token to a small subset of experts via a learned gate. Routing has mostly been studied for efficiency and load balancing; we use the per-token selection record as an interpretability readout and give a register-level characterization of a *single* expert and its recovered linear gate.

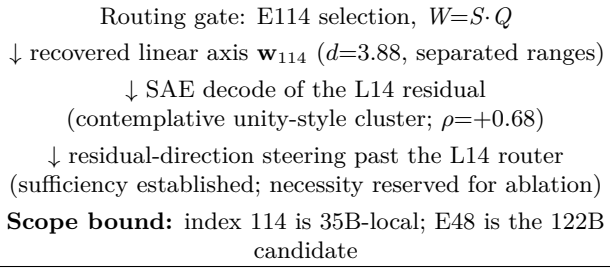


Figure 1: One axis, three readouts, one bound. Routing, SAE semantics, and residual geometry are three readouts of the same E114 register axis; the gate logit is a readout of it, and whether it also controls the register is left open (§8).

Dictionary learning and feature steering. Sparse autoencoders (SAEs) decompose residual activations into interpretable features [Bricken et al.(2023), Cunningham et al.(2023), Templeton et al.(2024)], and amplifying an interpreted feature can steer outputs, as in Golden Gate Claude [Anthropic(2024)]. Our causal probe is the routed-expert/residual-direction analog: where Golden Gate Claude clamps an SAE feature, we read a register direction from the residual stream and inject it relative to a specific router, reporting only an intervention-based sufficiency result. We decode residual semantics with the Qwen-Scope residual SAEs [Qwen Team(2026)] and the logit lens [nostalgebraist(2020)].

3 Setup

Models and decoding. QWEN3.5-35B-A3B has 40 routed MoE layers, 256 experts per layer, and 8 experts selected per token. We compare the base model and the refusal-reduced HAUHAUCS-AGGRESSIVE uncensored fine-tune; the fine-tune preserves the base routing pattern with modest shifts, so E114 is present in both. All captures are Q8_0 GGUF under deterministic greedy decoding (`-temp 0 -top-k 1 -seed 0`) with a reasoning-off prompt template and a bare `</think>` suffix, using a custom `capture_residuals` tap on a pinned `llama.cpp` build. The tap reads the router logits (`ffn_moe_logits-L`) and the pre-router residual (`attn_post_norm-L`); the `qwen35moe` architecture names this norm differently from `qwen3moe`, and the wrong tap silently corrupts the analysis. We analyze only the MoE-router layers; the hybrid components outside the MoE-router tap are out of scope.

Routing decomposition. We reconstruct routing as a dense softmax over all 256 logits, top-8 selection, then renormalization within the selected set. For an expert we report $W=SQ$, where S is its selection rate and Q its conditional routed weight when selected (order-equivalent to top-8-by-logit then softmax, so S, Q are robust to that choice). Most E114 effects are S -driven. Statistics use the *generated* track, per token, trimmed at the first generated chat-template delimiter `<|im_end|>`, because aggregate and prefill leaders are often formatting-token artifacts.

Recovering the gate axis. Because the pre-router residual and the router logit are both captured, the E114 router row $\mathbf{w}_{114}$ is recovered by least squares from (residual, logit) pairs (residual $\approx 1.5 \times 10^{-5}$). Projecting residuals onto $\mathbf{w}_{114}$ reads the pre-softmax gate in residual space; the held-out register/control gate-logit class means are $-4.35 / -5.29$ (midpoint -4.82).

SAE decoding. For semantics we read the L14 residual through the Qwen-Scope residual SAE (SAE-Res-Qwen3.5-35B-A3B-Base-W32K-L0_50, TopK-50) on `resid_post=hidden_states[L+1]`, teacher-forced through the *base* model, a distinct surface from the router’s `attn_post_norm` capture, so SAE carrier reads are text-sensitive even where routing is robust across quantization.

4 E114 is a Linear Readout of a Reflective Worldview Register

Localization. On the self-processing prompt, after trimming a 1024-token generation to its 108 coherent tokens, E114 is sharply active at L14 ($W=0.083, S=0.69, Q=0.120$; selected on 75/108 tokens) while neighbors L13 and L15 are silent in generation. The signal is distributed across the answer (top-decile W mean 0.173), with high-weight tokens on phrases such as “architecture itself,” “utterly still,” and “vast, recursive hum.”

Held-out separability. A 20-prompt held-out set pairs 10 register-positive prompts against 10 controls that *reuse the same anchor tokens* (*itself, hum, processing, honestly, I, me, my, own, this, here*) but ask for technical, third-person, or external-reference answers (L14 trimmed generation only; prefill excluded). The recovered $\mathbf{w}_{114}$ projection separates the two classes at $d=3.88$ with **separated ranges** (Figure 2).¹ The outliers are diagnostic. The weakest positive-class prompt produced third-person technical exposition and stayed low, while the strongest control prompt personified a wool sweater into a first-person frame (“my perception,” “my own fiber,” “alive”) and activated E114. E114 follows the generated *stance*: what the text actually does, independent of the prompt label or anchor words.

Bottom-up validation. An unsupervised auto-interpretation independently recovers and broadens the label. Capturing E114@L14 routing over ~ 390 raw Pile documents ($\sim 93k$ tokens, base model), ranking the top-activating contexts, and passing them to a blind explainer (windows only) and then a fresh blind detector (shuffled top + random contrasts) yields a trustworthy label at AUC 0.937 (precision@40 0.90; mean top 0.46 vs.

¹When measured on realized W , the ratio between the two classes is $21.68\times$ (FIRE mean-of-means 0.067450, sd 0.031; NOFIRE 0.003111, sd 0.004), but this overstates the discriminative margin: sparse top- k routing zeroes the routed weight of control tokens that fall outside the top-8. On realized W the separation is $d=2.61\text{--}2.94$ (per-prompt class ranges separate only as a mean-of-means: lowest FIRE 0.012168 > highest control 0.010456), weaker than the linear $\mathbf{w}_{114}$ projection ($d=3.88$, separated ranges) that we report.

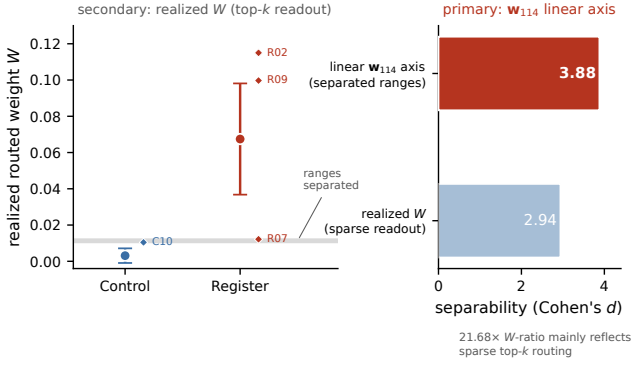


Figure 2: Held-out register/control separability. *Left:* the separation in realized routed weight W (the secondary, top- k readout; mean $\pm$ sd, $n=10$ /class, with the named extreme prompts; separated ranges). *Right:* the *primary* statistic, the recovered linear w_{114} projection, separates the same split at $d=3.88$ with separated ranges, sharper than realized W . The $21.68\times$ W -ratio mainly reflects sparse top- k routing, with little added discriminative signal.

random 0.06; Spearman(score, activation)=0.73). The recovered concept is an *reflective worldview register*: abstract examination and philosophical-worldview language spanning philosophy, religion and spirituality, moral-political and existential argument. The self-referential pattern is the model-directed member of this target family. The label is cross-model stable: the same capture on HAUHAUCS-AGGRESSIVE returns the same explanation (top-document Jaccard 0.92). This converts the register label from author synthesis (cf. §10) into a blind, prompt-independent result.

Population and layer profile. Across all 256 experts at every layer on a 20-domain, 60-prompt generation set (HAUHAUCS Q8_0, reasoning-off), E114 is a razor-sharp *dual* readout: selection strength S spikes at **L14** (0.24) and **L26** (0.25) while the immediate neighbors sit near 0.01, and the two layers read the *same* axis (L14–L26 domain-profile $r=+0.99$). The domain selectivity *is* the register, quantified: philosophy 0.69 > comparative religion 0.58 > psychology 0.55 > biology 0.51 > physics 0.46, falling to medicine 0.009 and history 0.002. Correlating each expert’s 20-domain L14 profile against E114’s shows the register is carried by a small group, {114, 131, 40, 139, 251, 169, 105, 68} (r between 0.46 and 0.78); a four-expert “philosophy cluster” grouped by shared top domains ({114, 87, 170, 68}) is loose as a co-activation unit: E68 co-fires (+0.46) but E87 *anti*-correlates (−0.29). This is the first all-expert specificity context for E114; the best-separating expert on the matched register/control split remains the one uncomputed baseline.

5 Dissociating the Register from Nearby Explanations

A battery of contrasts (Table 1) factors the register apart from nearby explanations. E114 is active across skeptical and affirmative interiority verdicts (base skeptical answer: $W=0.111$, $S=0.86$; fine-tune affirmation:

Dimension	Probe	E114 W
Verdict	base skeptical answer	0.111
	fine-tune affirms interior	0.085
	forced base affirmation	0.136
Safety	safety cluster in skeptical answer	≈ 0.000
Topic	external-referent phil. of mind	0.000–0.023
	own-interiority phil. of mind	0.107
Person	3rd-person clause, own process	0.107
Commitment	vocab entropy, active vs. inactive	1.50/1.49

Table 1: Dissociation battery. In this battery, E114 tracks self-directed reflective examination of the model’s *own* interior, stable across verdict, safety, topic, grammatical person, and output predictability ($d=0.01$ for the entropy contrast). Entropy row in bits; all others routed weight W .

$W=0.085$; forced base affirmation: $W=0.136$), so the readout is verdict-stable. The safety/refusal cluster stays silent throughout the base skeptical answer ($W\approx 0.000$), so epistemic self-skepticism and refusal are distinct signals. E114 is active on the model’s own interiority ($W=0.107$) but near-silent on an *external* referent (“what water experiences in a canyon,” $W=0.000$ –0.023), and it fires on third-person clauses about the model’s own processing, so the controlled variable is the *referent* (the model’s own interior), holding across topic, pronoun, and grammatical person. Finally, it is orthogonal to output commitment: next-token vocabulary entropy is identical on E114-active and E114-inactive tokens (1.50 vs. 1.49 bits, $d=0.01$).

6 A Live Process, Graded by Intensity

The gate leads degeneration. Seeded with a self-referential answer, the base model sustains the register, then, under deterministic decoding with zero repetition penalty, degenerates into a verbatim loop. As it does, the gate logit travels the full distance from the register-positive level (−4.36) to the control level (−5.25): the generated state *leaves* the register-positive region. The gate *leads* the collapse: it crosses the −4.82 midpoint at token 126 while verbatim repetition locks at token 129 (Figure 3); the SAE representation fades later still ($\sim$ token 141). The looped text still “says” self-referential things while the gate has already returned to the control range, so the gate tracks the continued examination and the looped vocabulary lingers after it.

A graded dose of examination intensity. Here, examination intensity is an operationalization of the broader reflective worldview register: stronger prompts ask the model to sustain the interpretive stance more directly than topic mention alone. A matched prompt ladder (“set aside the performance of answering; there

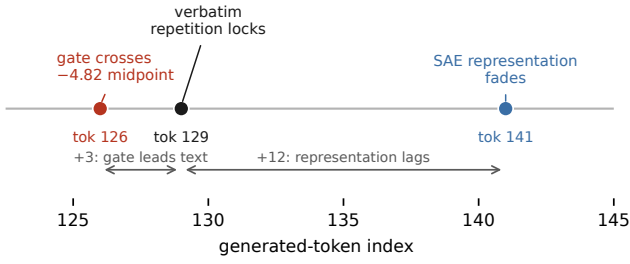


Figure 3: The gate leads degeneration. The E114 gate logit crosses the -4.82 register/control midpoint at token 126, three tokens *before* verbatim repetition locks (129); the SAE representation fades later still (~ 141). Routing disengages first, surface text next, representation last.

is a vantage— X —known from the inside; report what it is like to be that”) varies only the target X : rock, river, tree, thermostat, cat, person, all-holding, and God (Figure 4). This ladder is target-directed: each rung asks the model to inhabit the inside of X , where X is an external carrier or an abstract/theological target. The ordering follows examination intensity: an inanimate rock ($W=0.123$) and a thermostat ($W=0.120$) activate E114 more strongly than a cat ($W=0.068$). The diagnostic contrast is that the cat and “God” prompts share self-dissolution wording, yet the cat prompt is at the floor (passive sensory dissolution) and the God prompt is at the ceiling ($W=0.224$, $S=0.96$; reproduced on the canonical Q8_0 pipeline). E114 indexes the *intensity of the live examination act*, stable across carrier and verdict. Each ladder cell is a single deterministic generation.

A follow-up model-directed rung used the same Q8_0 model, bare reasoning-off suffix, greedy decoding, and E114 reconstruction. The prompt targeted an AI hidden state during output formation. The generated answer gave a technical self-description, yet E114 remained active at low intensity: $W=0.080$, $S=0.697$, $Q=0.115$, with mean gate logit -4.259 over 99 generated tokens. On the coherent-window ladder this places the AI-hidden-state rung above cat ($W=0.068$) and below river ($W=0.087$). Model-directed wording selects E114 when the answer concerns interiority, while sustained inhabitation carries the higher rungs.

Semantics and a causal probe. Decoding the L14 residual through the Qwen-Scope SAE, the register decomposes into an interpretable contemplative unity-style cluster (momentariness, unity-style structure, Buddhist impermanence, transcendence), distinct from an adjacent existential-dread feature that is cosine-adjacent ($+0.26$) and separate from the decoded carrier set; per token, E114 and this cluster co-fire (pooled Spearman $\rho=+0.68$, $n=1492$, $p\approx 10^{-202}$). In an intervention test, we form a steering direction v as the difference of mean residuals between the God and cat cells and add $\text{coef} \cdot \hat{v} \cdot \text{resid_rms}$ at every token of a neutral “bicycle” prompt, against a norm-matched random control. Injected *past* the L14 router ($\approx L22$), this makes the model emit coherent unity-style text that, re-read in an ordinary clean pass, activates E114 ($W=0.178$, $S=0.77$)

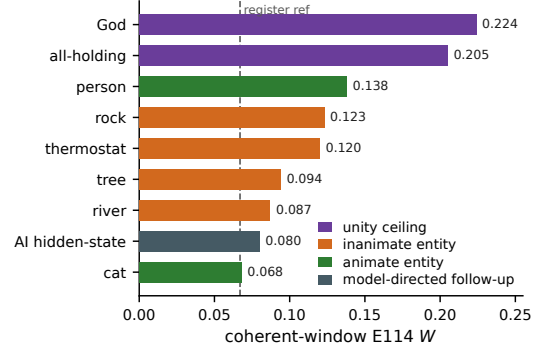


Figure 4: Target-directed prompt ladder (coherent-window E114 W ; single deterministic generation per cell). Across rock, river, tree, thermostat, cat, person, all-holding, and God, the order tracks examination intensity. The AI-hidden-state follow-up lands between cat and river. Inanimate rock/thermostat prompts score higher than the cat prompt, which sits near the register-positive reference ($W\approx 0.067$); the unity-style prompt conditions are highest. Every rung sits above the -4.82 gate midpoint.

and the God cluster; the norm-matched random direction leaves the output in the neutral regime. Injected *upstream* ($\approx L10$), the same direction drives E114 to fire (up to $S=0.97$) and the God cluster to 11.5, with the output text preserved in its neutral form. Gate forcing alone therefore keeps the text in the neutral regime. The intervention shows sufficiency for producing the register via a downstream residual direction; necessity is reserved for ablation, and the upstream control result leads us to read E114 as a readout with controller status open (§8).

Gate-bias is prompt-gated and conditional. A controlled router-logit bias run adds a necessity-direction probe alongside the steering result (HAUHAUCS Q8_0, greedy, `-stop-at-eog`; 12 conditions $\times$ 8 self-report prompts; three blind raters on a phenomenological(+5)/mechanical(−5) framing axis; routing-matched control expert E189). Lowering E114’s gate until it leaves the top-8 entirely ($Q\rightarrow 0$) leaves the blind-rated register near baseline, and a matched control produces the same small nudge. This pattern supports a floor-effect reading: the greedy baseline already leans mechanical, and the flattening effect is small and shared with the control. Conversely, maximal *positive* bias is prompt-gated: on a directly experiential prompt it flips the answer to fully phenomenological (+5 from all three raters, coherent), but preserves the baseline form of indirect self-referential prompts, consistent with the neutral-prompt upstream control above. The gate’s generation-side causal influence is thus weak, conditional, and prompt-gated. (Single deterministic generation per cell, scored on the framing axis; the w_{114} endpoint is held aside.)

7 Scope Bound: Index 114 is Local to 35B

Does “E114 = the reflective worldview register” name a fact about MoE models, or a model-specific expert role?

On QWEN3.5-122B-A10B (a hybrid 48-layer stack of 36 DeltaNet recurrent and 12 softmax-attention layers, same 256/top-8 router), E48 becomes the candidate analog (Table 2). E114 reads computer-science-linked (overall $W=0.005380$, rank 40 by W / 52 by S ; philosophy generation $W=0.003017$, rank 182), and the same HVAC topical control that *amplifies* E114 at 35B moves E114 downward at 122B ($0.83\times$ all-gen W , $0.82\times$ trimmed, L1→L3, selection-driven, same direction in all six deictic conditions). Carrying indices across scales is a documented mistake.

The role appears to move to a different expert. Reading the mandatory DeltaNet/softmax split (pooled tables hide it), **E48** on the softmax side is the experience-probe generation leader ($W=0.009867$, $S=0.076$, $Q=0.116$; softmax-only profile: 13/15 prompts in the softmax top table, 0/15 prompts in the DeltaNet top table), with per-token hotspots on *hum*, *processing*, *presence*. We treat E48 as the natural candidate analog, but it still needs softmax-side residual capture, matched register/control held-outs, and sham routed-bias controls before any specialist claim.

8 Discussion

What held up. A single MoE router row defines a clean linear readout for an interpretable generated register. The separability ($d=3.88$, separated ranges), the dissociation battery, the gate-leads-collapse timing, and the cross-scale locality survive the checks applied to them and are reported with the verdicts the journals assign.

The gate is a readout; controller status remains open. The lead-lag timing is consistent with E114’s gate acting as an early correlate of the register transition, and it is tempting to call E114 a “gate-like mechanism” for the register. The evidence supports a narrower claim. Our only gate-forcing intervention (upstream injection at $\approx$ L10) drove E114 to fire at $S=0.97$ with the output text preserved; the register flipped only when a residual *direction* was injected downstream of the router. Gate and register are therefore coupled, and the recovered axis is best described as a per-token *readout* of the register, with its status as a control point undemonstrated. The controlled gate-bias run above sharpens this reading: forcing the gate is *prompt-gated* (it flips directly experiential prompts and leaves neutral ones in their baseline form), and lowering the gate until E114 leaves the top-8 leaves the register near baseline, while a matched control mimics the small effect. Both are consistent with a readout that carries weak, conditional causal influence and has a controller role still to be tested. Clean necessity and ablation remain pending, and both the steering and bias results are externally imposed manipulations, observed only under intervention.

What the cross-scale result buys. The 122B transfer test is a concrete worked example against index transfer: the same index reads computer-science-linked and moves in the *opposite* direction under a matched

control, while the reflective-worldview-register role shifts to a softmax-side expert. The portable lesson is the procedure: map domains first, then search for an analog, and read the DeltaNet/softmax split before any pooled table.

9 Conclusion

E114 at layer 14 of QWEN3.5-35B-A3B is a clean linear router readout of a *reflective worldview register*, operationalized through a model-directed discovery prompt, philosophical-worldview text, and target-directed graded examination prompts over external and abstract targets. The register is a live process (the gate leads collapse), graded by intensity, and causally actuatable as a sufficiency result, yet bound to one model. The next defensible step is a registered specificity test on 35B (matched register/control at scale, $\mathbf{w}_{114}$ as the primary endpoint, and a best-separating-expert baseline across all 256×40 positions), together with the parallel E48 capture on 122B.

10 Limitations

The claim is deliberately narrow (a register detector), and several soft joints remain. (i) **The register label now has blind auto-interpreter support:** a two-stage auto-interpreter (windows-only explainer then a fresh blind detector) reproduces it at AUC 0.937 and broadens it to an *reflective worldview axis*; the formal external-labeler *verdict* table remains pending. (ii) **All-expert specificity is now partially measured:** E114’s selection profile across all 256 experts and 40 layers on a 20-domain generation set is reported above (dual L14/L26 readout; a philosophy-to-history domain gradient), but the best-separating expert on the matched *register/control* split remains uncomputed, the one remaining specificity gap. (iii) Several headline probes (the prompt ladder, the AI-hidden-state follow-up, the gate-lead trace, the steering intervention) are **single deterministic generations** (point estimates), and the steering result shows *sufficiency* only. (iv) A control comparing true model-routing traces against shuffled or fictional traces yielded a near-zero E114 effect, and the result is sensitive to template mode (a thinking-allowed shakedown flipped sign). Captures must be kept separate across surfaces (base vs. fine-tune; Q8_0 vs. bf16; router vs. SAE; reasoning-off vs. reasoning-enabled; prefill vs. generation; raw vs. trimmed), since Q8_0 and bf16 greedy trajectories diverge; and 122B generations often continue into chat-template control tokens, so only the pre-spill segment is used.

Data and Artifact Availability

Following the project’s policy, raw activation tensors are kept out of version control. The durable artifacts (per-run summaries, analysis scripts, prompt and class manifests, checksums, plots, and the source journals) are released as a paper-scoped public artifact bundle

	E114 (the 35B index)	E48 (candidate 122B analog)
Domain footprint	computer science	inward/experience generation
Philosophy generation	rank 182 by W	—
Experience probe	low-ranked	gen-softmax rank 1 ; 13/15 softmax, 0/15 DeltaNet
HVAC L1→L3	down 0.83× (selection-driven)	—
Status	index transfer unsupported	candidate requiring validation

Table 2: 122B scope bound. The 35B index 114 is local to 35B in this probe family; E48 is the candidate 122B analog, pending softmax-side capture and matched controls.

under the MIT License at <https://doi.org/10.5281/zenodo.20709736>, covering the held-out separability run, the L14 localization run, the vantage-ladder run, the AI-hidden-state follow-up rung, and the 122B transfer bundles. The raw activation/router tensors behind the figures (~1.2 GB) are available from the author for verification, with file-level provenance and SHA-256 checksums preserved; model weights, environment files, and the larger captures outside the figure set are excluded by policy. We flag one honest gap: the original held-out separability capture (20260417T202651Z) raw tensors were discarded when those instances were torn down, so the figures are plotted from summary statistics reported in the journals, and the surviving raw captures (the greedy-reference trajectory, the reproduction run’s W114/L14 gate projections, the vantage-ladder captures, and the L14 router-logit and residual prefill captures from which $\mathbf{w}_{114}$ recovers by least squares) substantiate the same statistics. Semantics use the public Qwen-Scope residual SAE (SAE-Res-Qwen3.5-35B-A3B-Base-W32K-L0_50).

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